Bank Customer Churn Analysis Using Python

An Exploratory and Statistical Investigation of Customer Attrition Patterns, Banking Behaviour, and Retention Opportunities

1. Project Overview

This project focuses on analysing bank customer data to understand customer churn patterns and identify characteristics associated with customers leaving the bank.

Using Python, we will examine customer attributes such as credit score, age, geography, gender, account balance, tenure, number of banking products, credit card ownership, active membership, and estimated salary.

The project will investigate which customer groups have higher churn rates, whether account activity and product usage are associated with customer retention, how churn differs across geographical regions, and which customer characteristics are worth further investigation.

The final goal is to transform raw customer records into meaningful statistical findings, visualizations, and evidence-based recommendations that may help banks improve customer retention.

The project will focus on exploratory data analysis and statistical investigation. Machine learning is not required.

2. Data Source

Dataset: Bank Customer Churn

Source: The CSV file selected for this project. The original source URL and dataset version will be documented once verified.

Detail	Description
Domain	Banking / Customer Analytics
File format	CSV
Main dataset	One customer-level CSV file
Expected attributes	13 columns, based on the supplied sample
Number of records	To be confirmed by inspecting the CSV
Analysis tool	Python

Environment	Google Colab or Jupyter Notebook

Dataset validation: After loading the file, we will use df.shape, df.columns, and df.info() to confirm the actual number of records, attributes, and data types. We will verify that the dataset contains at least 500 records and 10 features, as required by the project criteria.

The analysis will use the first CSV file containing 13 columns. The second file will not be merged unless its purpose and relationship to the primary dataset are established.

3. Problem Statement

Customer churn is an important concern for banks because losing customers can affect long-term customer relationships, product adoption, and business performance.

Although customer databases contain demographic, financial, and account-activity information, these attributes require systematic analysis to understand how customer churn varies across different groups.

This project aims to answer the following question:

Which customer characteristics and banking behaviours are associated with higher customer churn, and what retention opportunities can be identified from the available data?

The analysis will identify patterns and statistical associations. It will not claim that any particular characteristic causes a customer to leave the bank.

4. Project Objectives

1. Understand the dataset structure, attributes, and data quality.
2. Clean missing values, duplicates, inconsistent formats, and incorrect data types.
3. Calculate descriptive statistics for numerical customer attributes.
4. Examine the overall distribution of customer churn.
5. Compare churn rates across age groups, genders, and geographical regions.
6. Investigate the relationship between credit scores and customer churn.
7. Analyse churn patterns by account balance and estimated salary.
8. Compare churn between active and inactive bank members.
9. Investigate churn by tenure, credit card ownership, and number of products.
10. Examine relationships between numerical customer attributes using correlation analysis.

11. Develop at least 15 meaningful visualizations.
12. Derive at least five evidence-based findings and propose practical customer-retention recommendations.

5. Attribute (Column / Feature) Details

The following attributes are based on the sample supplied. Their exact data types and values will be verified after loading the CSV.

Attribute	Description	Analytical purpose
CustomerId	Unique customer identifier	Record validation and uniqueness checks
Surname	Customer surname	Data-quality checks; generally excluded from statistical comparisons
CreditScore	Customer credit score	Investigate credit-score differences by churn status
Geography	Customer's country or region	Compare regional churn rates
Gender	Customer gender	Compare churn rates between recorded groups
Age	Customer age	Analyse age distributions and churn patterns
Tenure	Number of years with the bank	Investigate customer loyalty and retention
Balance	Account balance	Compare balances by churn status
NumOfProducts	Number of banking products held	Analyse product usage and churn
HasCrCard	Credit card ownership indicator	Compare churn among cardholders and non-cardholders
IsActiveMember	Active membership indicator	Investigate the association between account activity and churn
EstimatedSalary	Estimated customer salary	Examine salary distributions and churn

Exited	Customer churn indicator	Main outcome variable for descriptive comparisons

Target variable: Exited

- 1 = Customer exited the bank.
- 0 = Customer did not exit the bank.

We will verify that these meanings are consistent with the dataset documentation.

6. Tools and Technologies

Technology	**Usage**
Python	Complete analytical workflow
Pandas	Data loading, cleaning, grouping, and summary tables
NumPy	Numerical calculations and derived features
Matplotlib	Histograms, bar charts, and line charts
Seaborn	Box plots, count plots, and correlation heatmaps
Jupyter Notebook / Google Colab	Executable Python notebook
GitHub	Project versioning, documentation, and portfolio presentation

The project will be completed entirely in Python. Excel, SQL, and Power BI will not be required.

7. Data Preprocessing and Feature Engineering

A. Data Cleaning

We will:

- Load the CSV using pd.read_csv().
- Inspect the first and last rows using head() and tail().
- Check dataset dimensions using shape.
- Review column names and data types.
- Calculate missing values using isnull().sum().
- Identify complete duplicate rows.

- Check whether CustomerId values are unique.
- Examine categorical values for inconsistent spelling or formatting.
- Verify numerical columns for unexpected values and incorrect types.
- Check the validity of Exited, HasCrCard, and IsActiveMember.
- Investigate unusual values in age, credit score, balance, tenure, and salary.
- Preserve legitimate zero balances rather than treating them as missing values.
- Document every cleaning decision.

We will not remove records or outliers automatically. Each decision will be based on data quality and analytical relevance.

B. Feature Engineering

Derived column	Method	Purpose
Age_Group	Group ages into documented age bands	Compare churn across age groups
Tenure_Group	Group customers by years with the bank	Examine retention by customer tenure
CreditScore_Group	Create documented credit-score bands	Compare churn across score ranges
Balance_Group	Categorize account balances	Compare churn by balance category
Salary_Group	Categorize estimated salary	Explore churn patterns across salary bands
Churn_Status	Map Exited to descriptive labels	Make charts easier to interpret
Active_Status	Map activity indicator to descriptive labels	Compare active and inactive members
CreditCard_Status	Map card indicator to descriptive labels	Compare cardholders and non-cardholders

The original columns will be retained. Any categories will be defined consistently and documented before interpreting the results.

8. Analysis and Visualizations

The exploratory data analysis will be organized into three categories:

- 5 univariate visualizations
- 5 bivariate visualizations
- 5 multivariate visualizations

This provides a plan for at least 15 charts. The final chart selection may be adjusted if particular variables contain insufficient variation or data quality issues.

9. Exploratory Data Analysis (EDA)

Before visualization, we will calculate:

- Total customer records.
- Number of unique customer IDs.
- Total customers who exited and remained.
- Overall churn percentage.
- Missing-value percentage for every column.
- Number of duplicate records.
- Distribution of customer ages and credit scores.
- Average and median account balances.
- Distribution of tenure and number of banking products.
- Customer counts by geography and gender.
- Churn counts and rates across important customer groups.

These summaries will help establish the dataset's structure and determine which comparisons are meaningful.

10. Statistical Analysis

Statistic or method	**Application**
Mean	Average age, credit score, balance, and salary
Median	Typical values less affected by extreme observations
Mode	Most frequent values or categories
Minimum and maximum	Identify value ranges
Standard deviation	Measure numerical variability

Variance	Quantify dispersion
Quartiles and IQR	Understand spread and investigate potential outliers
Frequency distributions	Summarize customer categories
GroupBy summaries	Compare churn across customer groups
Pivot tables	Compare customer characteristics by churn status
Churn rate calculations	Calculate the proportion of customers who exited
Pearson correlation	Explore linear relationships between numerical variables
Spearman correlation	Explore monotonic relationships where appropriate

Churn rates will be calculated as the number of customers who exited divided by the total number of customers in the relevant group, multiplied by 100.

Correlation will be interpreted carefully. It does not establish causation, and the binary churn indicator will not be treated as an ordinary continuous measurement without qualification.

11. Data Visualization

The following is the planned chart list. Actual charts, numerical results, and interpretations will be generated from the CSV.

A. Univariate Analysis — 5 Charts

1. Customer Churn Distribution — Count Plot

What we will do: Count customers who exited and those who remained.

Interpretation: Establish the class distribution and the overall scale of customer attrition.

2. Customer Age Distribution — Histogram

What we will do: Visualize the distribution of customer ages.

Interpretation: Identify common age ranges and potential age-related patterns worth investigating.

3. Credit Score Distribution — Histogram

What we will do: Examine the spread of customer credit scores.

Interpretation: Understand the typical score range and identify unusual observations.

4. Account Balance Distribution — Histogram

What we will do: Examine account balances, including the number of zero-balance accounts.

Interpretation: Determine how customer balances are distributed and whether the distribution is strongly skewed.

5. Customer Count by Geography — Bar Chart

What we will do: Count customers in each geographical group.

Interpretation: Understand the geographical composition of the dataset before comparing regional churn rates.

B. Bivariate Analysis — 5 Charts

6. Age Group vs Churn Rate — Bar Chart

What we will do: Calculate churn rates for defined age groups.

Interpretation: Identify age groups with comparatively high or low observed churn rates.

7. Geography vs Churn Rate — Bar Chart

What we will do: Calculate the percentage of customers who exited in each geographical group.

Interpretation: Compare customer attrition across regions while considering the number of customers in each group.

8. Active Membership vs Churn Rate — Bar Chart

What we will do: Compare churn rates for active and inactive members.

Interpretation: Investigate whether inactive membership is associated with higher observed churn.

9. Number of Products vs Churn Rate — Bar Chart

What we will do: Calculate churn rates for customers holding different numbers of products.

Interpretation: Examine whether product holdings are associated with customer retention.

10. Credit Score vs Churn Status — Box Plot

What we will do: Compare credit-score distributions for customers who exited and those who remained.

Interpretation: Assess differences in medians, spread, and overlapping distributions.

C. Multivariate Analysis — 5 Charts

11. Numerical Feature Correlation — Heatmap

What we will do: Calculate correlations among age, credit score, tenure, balance, number of products, and estimated salary.

Interpretation: Identify numerical relationships that warrant further investigation.

12. Age Group and Geography vs Churn Rate — Grouped Bar Chart

What we will do: Compare age-group churn rates across geographical groups.

Interpretation: Examine whether age-related churn patterns differ by region.

13. Balance vs Estimated Salary by Churn Status — Scatter Plot

What we will do: Plot account balance against estimated salary, distinguishing customers by churn status.

Interpretation: Explore the joint distribution of financial attributes and churn groups.

14. Tenure Group and Active Membership vs Churn Rate — Grouped Bar Chart

What we will do: Compare churn rates across tenure groups for active and inactive customers.

Interpretation: Examine whether the relationship between tenure and churn differs by membership activity.

15. Customer Profile vs Churn — Stacked Bar Chart

What we will do: Compare churn status across combinations of product count and credit card ownership.

Interpretation: Identify customer groups with different observed churn distributions.

All charts will be generated using Python. Their interpretation will be based on calculated results rather than assumptions about what the charts will show.

12. Summary of Findings

The completed project will present at least five findings supported by calculations and visualizations.

The investigation will focus on:

1. Overall churn: What proportion of customers exited?
2. Demographic patterns: Which age groups and geographical groups have higher observed churn rates?
3. Account activity: How does churn differ between active and inactive members?

4. Product usage: How do churn rates vary with the number of products held?
5. Financial characteristics: How do balance and credit-score distributions compare between churn groups?
6. Customer tenure: Are churn rates different among newer and longer-standing customers?
7. Data quality: Are there missing values, duplicates, or unusual records that affect interpretation?

Actual counts, percentages, comparisons, and chart references will be reported after running the analysis. No findings will be invented in advance.

13. Types of Analysis

Analysis type	**Planned work**
Descriptive	Summarize customer characteristics, churn counts, churn rates, and numerical distributions.
Diagnostic	Investigate differences in churn across demographics, financial attributes, account activity, and product usage.
Predictive	Not included in the core project; machine learning may be considered as a future extension.
Prescriptive	Recommend customer-retention investigations based on observed patterns and the limitations of the dataset.

The project will identify associations, not prove why a customer left the bank. Recommendations will therefore be framed as opportunities for further investigation rather than guaranteed solutions.

14. Future Enhancements and Suggestions

Potential future extensions include:

- Adding additional customer transaction or service-interaction data.
- Investigating churn over time if dated customer records become available.
- Conducting statistical significance tests for selected group comparisons.
- Building a churn-prediction model using scikit-learn.
- Exploring explainability methods for a future predictive model.
- Automating data-quality checks and periodic analytical reports.

- Extending the analysis to a larger, verified banking dataset.

These enhancements are optional and are not required to complete the current Python project.

15. Conclusion

This project will demonstrate a complete Python data analytics workflow using a bank customer churn dataset.

Through data cleaning, exploratory data analysis, statistical summaries, and 15 planned visualizations, the study will investigate customer attrition patterns across demographic, financial, and banking-behaviour attributes.

The final report will present actual findings and evidence-based recommendations while acknowledging the limits of the available data.

The completed project will demonstrate practical skills in Python, Pandas, NumPy, Matplotlib, Seaborn, statistical analysis, and data storytelling, making it a useful addition to a data analytics portfolio.